

Sector Rotation Screener

Methodology & Logic — how the scanner ranks NSE Nifty 500 stocks

EOD screener • RRG + Relative Strength + 200-DMA Trend + Sector Breadth

This document explains, end to end, how the screener decides which stocks and sectors look strong. It is a transparent record of the exact inputs, formulas, weights and limitations — so the output can be understood and trusted, not taken on faith.

1. What the scanner does (the core idea)

At any time, capital in the market rotates — flowing out of some sectors and into others. A stock can have great fundamentals yet go nowhere if its sector is out of favour. The scanner is built around this idea of **rotation**: find where strength is flowing *now*, then pick the healthiest stocks inside that strength.

It scans every stock in the NSE Nifty 500 and blends four complementary checks into a single **Rotation Score from 0 to 100**. Each check covers a blind spot of the others:

- **RRG** tells you the rotation picture — but it is purely *relative*, so a sector can look good while still falling.
- **Relative Strength ranking** turns that into a hard, comparable number across the universe.
- **The 200-DMA trend filter** removes the trap of "leading in a downtrend" — it keeps only stocks actually trending up.
- **Sector breadth** confirms the move is real participation, not one lucky stock carrying a sector.

Used together, they answer one question: *which stocks are strong, in an uptrend, and inside a participating sector — right now?*

2. The data it runs on

- **Universe**: the official NSE Nifty 500, grouped by NSE's own "Industry" classification into 20 macro-sectors. Refreshable from the published constituent list.
- **Prices**: daily end-of-day (EOD), **split/dividend-adjusted**, so trends and returns are continuous through corporate actions.
- **History**: about 7 years per stock — enough for the longer timeframes to compute.
- **Benchmark**: NIFTY 500 by default (configurable to NIFTY 50 or MIDCAP 150).
- **Integrity**: prices are cross-validated between two independent sources before use (see Section 6).

3. The four analytical layers

3.1 Layer 1 — Relative Rotation Graph (RRG)

RRG measures a stock (or sector) against the benchmark on two axes: **RS-Ratio** (is it outperforming?) and **RS-Momentum** (is that outperformance improving?). Both are normalised around 100, so 100 = in line with the benchmark.

```
raw_rs = (stock_close / benchmark_close) * 100
rs_ratio = 100 + (raw_rs - SMA(raw_rs, W)) / StdDev(raw_rs, W)
roc = rs_ratio / rs_ratio.shift(L)
rs_momentum = 100 + (roc - SMA(roc, W)) / StdDev(roc, W)
```

W is the smoothing window and L the momentum look-back (both set per timeframe, Section 5). The (RS-Ratio, RS-Momentum) pair places the name in one of four quadrants, and the direction is

"accelerating" if the latest momentum is rising, else "decelerating":

Quadrant	RS-Ratio	RS-Momentum	Meaning
Leading	> 100	> 100	strong and still gaining
Improving	< 100	> 100	weak but turning up
Weakening	> 100	< 100	strong but losing steam
Lagging	< 100	< 100	weak and getting weaker

3.2 Layer 2 — Relative Strength (RS) ranking

For each stock, the excess return over the benchmark is measured across several look-backs, combined into a weighted composite, then rank-converted to a percentile (0-100) across the universe. Recent look-backs are weighted more heavily.

```
for each look-back: RS = (stock_return) - (benchmark_return)
composite_rs = sum( weight_i * RS_i ) weights = 0.15, 0.25, 0.30, 0.30
percentile = rank of composite_rs across the universe, scaled 0-100
```

Note: in the per-sector drill-down the percentile is computed **within that sector**; in the full scan it is computed across the whole universe.

3.3 Layer 3 — Absolute trend filter (200-DMA)

A purely relative signal can flag a stock that is falling, just falling less than the index. This filter keeps only stocks trading above their 200-day moving average — a simple, widely-used definition of a long-term uptrend.

```
above_200dma = close > SMA(close, 200)
trend_score = 100 if above else 0
pct_from_dma = (close / SMA(close, 200) - 1) * 100 ( near if within +/- 2% )
```

3.4 Layer 4 — Sector breadth

Breadth checks how much of a sector is participating — the share of its constituents above their 50- and 200-day moving averages. Broad participation = healthier, more durable move.

```
breadth_N = (constituents above their N-DMA) / total * 100 for N = 50 and 200
signal: >= 70 Strong | 40 to 69 Mixed | < 40 Weak
```

4. The composite Rotation Score (0-100)

The four layers are blended into one score. Relative strength carries the most weight; the RRG quadrant, trend and breadth round it out:

```
rotation_score = 0.25 * RRG_score + 0.35 * RS_percentile + 0.20 * trend_score + 0.20 * breadth_200
```

The RRG quadrant + direction maps to an RRG_score as follows (higher = better placed in the rotation cycle):

Quadrant / direction	RRG_score
Leading + accelerating	100
Improving + accelerating	80
Leading + decelerating	70

Weakening + accelerating	50
Improving + decelerating	40
Weakening + decelerating	30
Lagging + accelerating	20
Lagging + decelerating	10

If a component cannot be computed (e.g. too little history), it falls back to a neutral 50 so the score stays defined — worth remembering when judging stocks with short histories.

5. Timeframes

The same engine runs on five timeframes. Shorter = more responsive; longer = smoother, fewer signals. W is the RRG smoothing window, L the momentum look-back (on resampled bars).

Timeframe	Bars	W (smoothing)	L (momentum)
Weekly	weekly	14	52
Monthly	monthly	10	12
Quarterly	quarterly	4	4
Semi-annual	2 x quarter	3	2
Annual	yearly	2	1

Longer timeframes need many years of data to compute; for newer listings they may be unavailable and are shown as N/A rather than guessed.

6. Data-integrity engine (why the numbers can be trusted)

Bad price data silently corrupts every indicator. To guard against it, prices are reconciled across two independent sources before they reach the scanner:

- **Two sources:** an adjusted feed (the analytical series) is cross-checked against the official NSE feed.
- **Return-level validation:** the two are compared on daily returns (which match across sources except on corporate-action days), not raw price levels.
- **Bad-tick repair:** isolated one-day spikes that reverse the next day are detected and repaired by interpolation.
- **Quality status per stock:** CONFIRMED (sources agree), PARTIAL, SUSPECT (disagree — review before official use), or SINGLE-SOURCE (only one feed available).

Every stock carries its quality status in the tool, so you always know how well-validated a given name's data is.

7. How to read the output

- **Quadrant** — Leading / Improving / Weakening / Lagging (see 3.1).
- **RS-Ratio > 100** = beating the index; **RS-Mom > 100** = that lead is growing.
- **200-DMA** — a tick means the stock is in a long-term uptrend; the % is distance from it.
- **Breadth** — Strong / Mixed / Weak participation of the sector.
- **Score** — the blended 0-100 rank; higher = a stronger, healthier setup now.

8. Limitations & disclaimer

- It is a **screenener, not a predictor**: it describes what looks strong now, it does not forecast prices or guarantee returns. Momentum can reverse sharply.
- It ignores valuation, fundamentals, news, liquidity, position sizing and taxes.
- The universe is today's Nifty 500 (current membership).
- Data is end-of-day and may lag by a trading day; it is not real-time.

Not investment advice. This tool and document are for education and research only. Do your own due diligence and consider consulting a SEBI-registered investment adviser before making investment decisions.